

Unit 5, Lesson 4: Exploring Corresponding Parts of Congruent Triangles are Congruent



Benchmarks

MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.

MA.912.GR.2.3 Specify a sequence of transformations that will map a given figure onto itself, or onto another congruent or similar figure.

MA.912.GR.2.6 Apply rigid transformations to map one figure onto another to justify that the two figures are congruent.



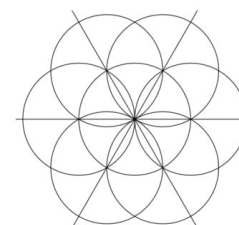
Learning Targets

- ☐ I can relate the definition of congruence in terms of rigid motions to corresponding parts of congruent triangles.
- ☐ I can use corresponding parts of congruent triangles to write congruence statements.



5.4.1 Warm-Up: Art in Geometry

Salih is taking an art class that looks at the different patterns in tiles from across the world. Salih used a compass to create this first step in a 12-fold pattern.



1. What do you notice about the design?

The last step of the 12-fold pattern and the final design are shown.

Last Step	Final Design

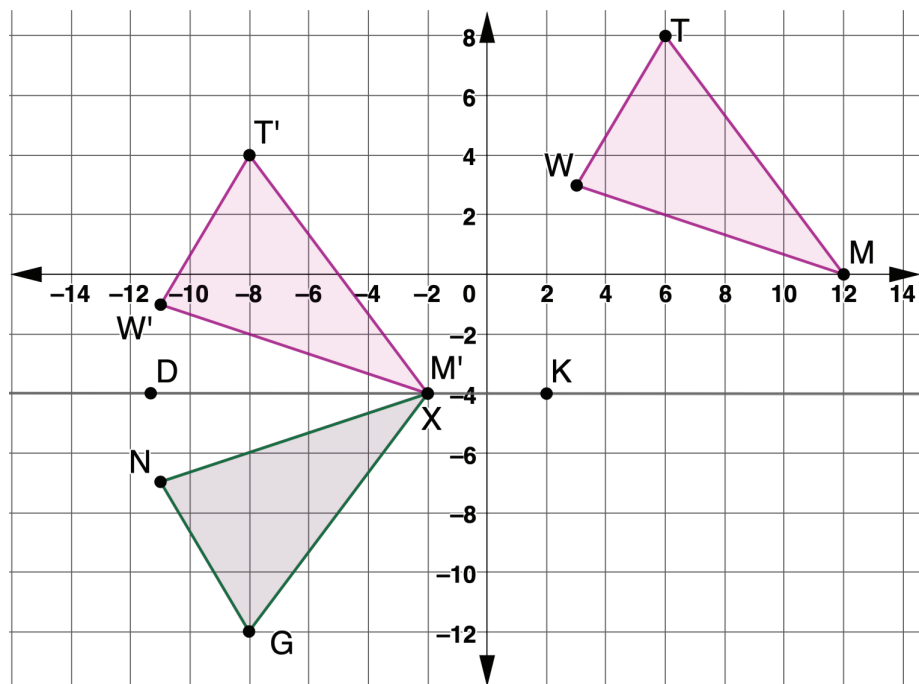
2. What do you wonder about the pattern or the final design?



5.4.2 Collaboration: Congruence in Terms of Rigid Motions

Work with your partner on the following.

The diagram shows a sequence of transformations that, when applied to $\triangle TMW$, show that $\triangle TMW$ maps to $\triangle GXN$.



1. What are the transformations used to map $\triangle TMW$ to $\triangle GXN$?

2. Complete the table by determining which parts of $\triangle GXN$ correspond with the given part of $\triangle TMW$ after mapping $\triangle TMW$ to $\triangle GXN$.

$\triangle TMW$	$\triangle GXN$
$\overline{MT}$	
$\overline{MW}$	
$\overline{WT}$	
$\angle TMW$	
$\angle TWM$	
$\angle WTM$	



Definition of Congruence in Terms of Rigid Motions

Two figures are congruent if and only if there exists one or more rigid motions, which will map one figure onto the other.

3. Discuss with your partner how to determine if the transformations that were used to map $\triangle TMW$ to $\triangle GXN$ are rigid motions.
4. Complete the statement.

$\triangle TMW$ and $\triangle GXN$ are congruent because there exists

- ☐ 1
☐ 2
☐ 3

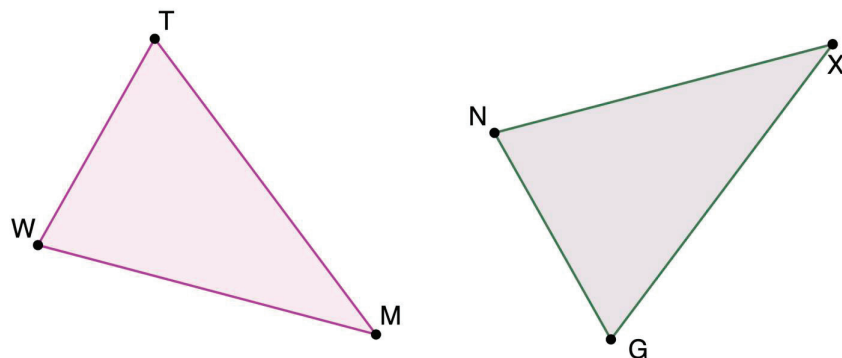
rigid motion(s) which map(s) $\triangle TMW$ to $\triangle GXN$.



5.4.3 Guided Instruction: CPCTC

Another way to show that triangles are congruent is with congruence postulates and theorems.

1. $\triangle TMW$ and $\triangle GXN$ are shown where $\angle GNX \cong \angle MWT$, $\angle NXG \cong \angle TMW$, and $\overline{XG} \cong \overline{MT}$.



- a. From the given information, what congruence postulate or theorem can be used to show that $\triangle TMW \cong \triangle GXN$?
- b. Discuss with your partner what could be concluded about the corresponding parts of the triangles if the congruence postulate or theorem is used to show the triangles are congruent.

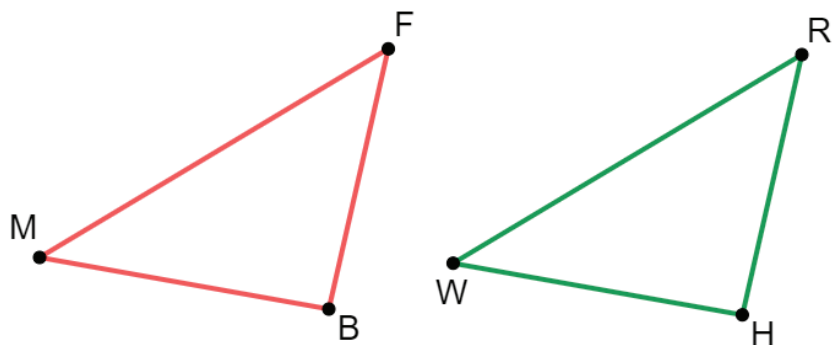


Corresponding Parts of Congruent Triangles are Congruent (CPCTC)

If two triangles are congruent, then all corresponding parts of the triangles are congruent.

There is a difference between naming two triangles and stating that two triangles are congruent. Order is important in a congruence statement, but is not necessary when only naming two triangles. The order used for congruence statements communicates which parts of the triangles correspond.

2. A diagram of two triangles is shown.

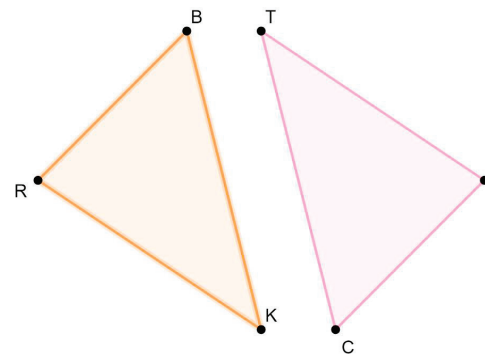


Naming	Congruence
ΔMFB and ΔHRW	$\Delta MFB \cong \Delta WRH$

Use the congruence statement to name all of the corresponding parts of the triangles.

ΔMFB	ΔWRH

3. Consider triangles BKR and CTY .



- If $\Delta BKR \cong \Delta CTY$, mark the corresponding congruent sides and the corresponding congruent angles on the diagram.
- Complete each congruence statement.

$$\overline{BK} \cong \underline{\hspace{2cm}} \quad \angle KBR \cong \angle \underline{\hspace{2cm}}$$

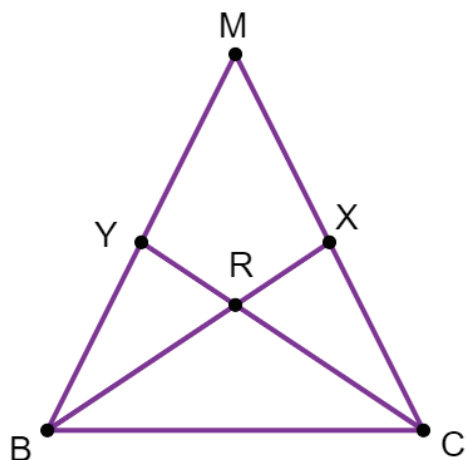
$$\underline{\hspace{2cm}} \cong \overline{CY} \quad \angle RKB \cong \angle \underline{\hspace{2cm}}$$

$$\overline{RK} \cong \underline{\hspace{2cm}} \quad \angle \underline{\hspace{2cm}} \cong \angle CYT$$

4. Consider two congruent triangles, ΔGRD and ΔFHW .

- Write the congruence statement $\Delta GRD \cong \Delta FHW$ in two other ways.
- Name the congruent sides for $\Delta GRD \cong \Delta FHW$.
- Name the congruent angles for $\Delta GRD \cong \Delta FHW$.

5. Triangle MBC is shown.



a. How many different triangles are in $\triangle MBC$?

b. If $\triangle BRY \cong \triangle CRX$, then complete the congruence statement.

$$\overline{RC} \cong \underline{\hspace{1cm}} \quad \angle RYB \cong \underline{\hspace{1cm}}$$

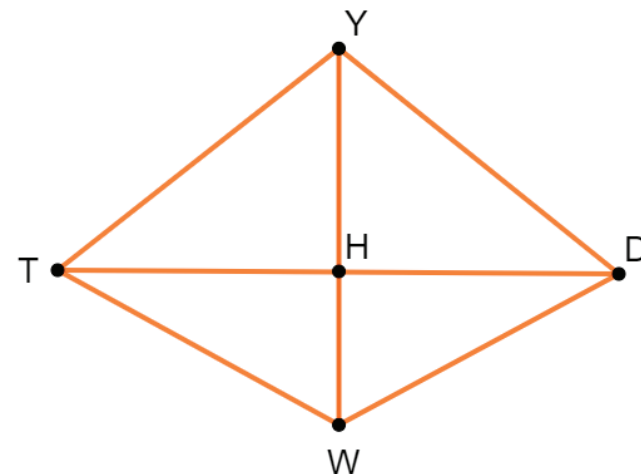
c. If $\triangle BXM \cong \triangle CYM$, then complete the congruence statement.

$$\overline{CY} \cong \underline{\hspace{1cm}} \quad \overline{CM} \cong \underline{\hspace{1cm}} \quad \angle MCY \cong \underline{\hspace{1cm}}$$



5.4.4 Guided Practice: Using CPCTC

Quadrilateral $YDWT$ is shown where $\overline{YH}$ is a perpendicular bisector of $\overline{TD}$.



1. How many different triangles are in Quadrilateral $YDWT$?

2. If $\triangle DHY \cong \triangle THY$, then complete each congruence statement.

$$\overline{HD} \cong \underline{\hspace{1cm}} \quad \angle DYH \cong \underline{\hspace{1cm}} \quad \overline{TH} \cong \underline{\hspace{1cm}}$$

3. If $\triangle DHW \cong \triangle THW$, then complete each congruence statement.

$$\angle TWH \cong \underline{\hspace{1cm}} \quad \angle HTW \cong \underline{\hspace{1cm}} \quad \overline{TW} \cong \underline{\hspace{1cm}}$$

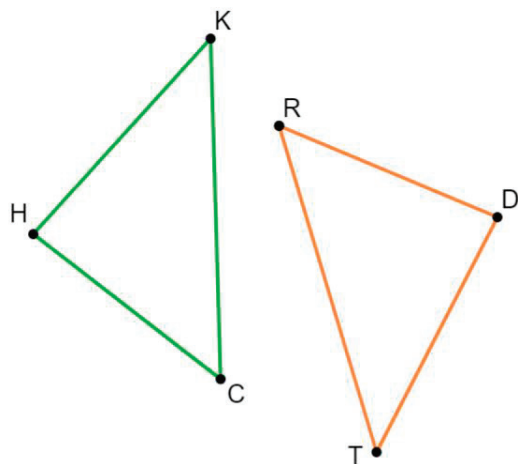
4. If $\triangle YTW \cong \triangle YDW$, then complete each congruence statement.

$$\angle WTY \cong \underline{\hspace{1cm}} \quad \overline{DW} \cong \underline{\hspace{1cm}} \quad \overline{YW} \cong \underline{\hspace{1cm}}$$



5.4.5 Wrap-Up: Cool Down

The diagram shows $\triangle HKC$ and $\triangle TDR$ where $\triangle HKC \cong \triangle TDR$.



1. Complete each congruence statement.

$$\overline{KC} \cong \underline{\hspace{2cm}}$$

$$\overline{CH} \cong \underline{\hspace{2cm}}$$

$$\angle RTD \cong \underline{\hspace{2cm}}$$

$$\angle RDT \cong \underline{\hspace{2cm}}$$

2. Becky told her teacher that there was a typo in the first sentence. She circled the typo.

The diagram shows $\triangle HKC$ and $\triangle TDR$ where $\triangle HKC \cong \triangle TDR$.

Write a note to Becky explaining why this is NOT a typo.

Unit 5, Lesson 5: Drawing Conclusions from Given Information



Benchmarks

MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Targets

- ☐ I can draw and mark figures using given information.
- ☐ I can use definitions, properties, postulates, and theorems to draw conclusions about triangles from given information.